

Zhaoyi Sun

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EDUCATION

University of Washington

Doctor of Philosophy in Biomedical and Health Informatics | GPA: 3.85/4.0

Seattle, WA

June 2028 (Expected)

Weill Cornell Medical College, Cornell University

Master of Science in Health Informatics | GPA: 4.0/4.0

New York, NY

July 2022

Nanjing University

Bachelor of Science in Chemistry | GPA: 3.62/4.0

Nanjing, China

June 2019

RESEARCH EXPERIENCE

University of Washington

Research Assistant

Seattle, WA

September 2023 – present

- Under the guidance of Prof. Meliha Yetisgen, focused on applying NLP and LLMs on information extraction and question answering tasks
- Collaborated with Microsoft on creating the benchmark datasets for MediQA shared tasks
- Working on creating a database of radiology incidental findings including related concepts, image follow-up status, and associated cancer related outcomes
- Working on applying LLM agents on multimodal error detection in abdominal CT reports
- Published a review on medically inaccurate information, including errors, misinformation and hallucination

Research Assistant

March 2025 – present

- Under the guidance of Prof. Trevor Cohen, focused on applying NLP and LLMs to solve mental health problems
- Developed LLM systems for classifying behavioral stages on QuitNet data
- Working on building LLM systems for depression detection on social media data

Boston Children's Hospital, Harvard Medical School

Intern

Boston, MA

June 2025 – September 2025

- Under the guidance of Prof. Tim Miller, focused on applying NLP and LLMs on biomedical regulatory text
- Working on creating a dataset of eligibility criteria edits for the clinical trials on ClinicalTrials.gov
- Developed LLM systems for classifying relations between medical device and adverse events on the International Medical Device Regulators Forum (IMDRF) data

Weill Cornell Medical College, Cornell University

Research Assistant

New York, NY

July 2022 – August 2023

- Under the guidance of Prof. Yifan Peng, focused on developing and applying computational approaches to biomedical text data and medical images, and evaluating large language models on clinical tasks
- Evaluated the performance of large language models on medical evidence summarization (GPT-3.5 and GPT-4) and impressions generation of radiology reports (GPT-4)
- Developed NLP pipelines to identify online health education materials to prevent postpartum depression using feature-based methods (TF-IDF and Random Forest), deep-learning-based methods (BERT), and ChatGPT
- Adapted a suicide-specific SDoH ontology and used NLP to identify individual-level SDoH-related social risks from death investigation narratives
- Published a scoping review on multi-modality joint learning in biomedical images and texts

Graduate Research Assistant

December 2021 – July 2022

- Under the guidance of Prof. Thomas Campion, focused on extracting structured and unstructured information from EHRs to support cancer clinical trials
- Conducted manual annotation of patients' ECOG/KPS scores and BRAF/KRAS mutation status in clinical notes in WCM's EHR system using REDcap

- Developed NLP pipelines using Python to automate data extraction from over 300 physician notes and genomic test reports for candidate identification for cancer clinical trials

Capstone Project Manager

October 2021 – July 2022

- Designed algorithms using Python for multiple-label classification of biomedical texts concerning hallmarks of cancer with deep learning
- Performed statistical analyses using R for predictors of skeletal-related events among bladder cancer patients with bone metastases
- Investigated the impact of built environments and social determinants on people's health status in NYC using SQL, Access, and Python
- Explored the contribution of imaging standards (DICOM and PACS) to healthcare interoperability and computer-aided diagnosis

Institute of Drug R&D, Medical School, Nanjing University

Nanjing, China

Research Assistant

July 2017 – January 2020

- Under the guidance of Prof. Yiqiao Hu and Prof. Jinhui Wu, focused on the combination of radiotherapy and immunotherapy for treating tumors and smart drug delivery systems
- Investigated the roles of DNA damage and STING pathway in tumor immunotherapy
- Conducted cell culture, animal experiments, organic synthesis, and immune characterization (ELISA and Flow Cytometer)
- Processed animal data for visualization using GraphPad as well as performed medical statistics analyses
- Published several papers in SCI core journals as the first author and co-author, one of them has been cited nearly 200 times in the last two years

Institute of Computing Technology, Chinese Academy of Sciences

Beijing, China

Data Analyst Internship

October 2018

- Constructed a peptide sequence search engine based on MATLAB and mass spectra
- Led a team to review literature about bioinformatics, genomics, and proteomics and prepared a systematic literature review

School of Chemistry and Chemical Engineering, Nanjing University

Nanjing, China

Research Assistant

March 2017 – July 2017

- Explored the methodology of gold-catalyzed organic synthesis
- Worked with Prof. Shaozhong Wang, prepared a series of alkynol compounds, and characterized them by ^1H NMR and infrared spectroscopy (IR)

TEACHING AND PROFESSIONAL EXPERIENCE

Weill Cornell Medical College, Cornell University

New York, NY

Teaching Assistant

December 2022 – April 2023

- Worked as a teaching assistant for the Natural Language Processing course, performed duties including mentoring, lecturing, researching, and grading for 30+ students

HONORS AND AWARDS

- **Certificate of Academic Excellence**, Weill Cornell Graduate School of Medical Sciences *July 2022*
- **The People's Scholarship in China**, Nanjing University *November 2018*
- **The People's Scholarship in China**, Nanjing University *November 2017*

PUBLICATIONS

Journal Articles

- **Sun Z**, *et al.* A scoping review of natural language processing in addressing medically inaccurate information: Errors, misinformation, and hallucination. *Journal of Biomedical Informatics* 2025, 104866
- Lin M, Li T, **Sun Z**, *et al.* Improving Fairness of Automated Chest X-ray Diagnosis by Contrastive Learning. *Radiology: Artificial Intelligence*, 6(5), e230342

- Patra BG, **Sun Z**, *et al.* Automated classification of lay health articles using natural language processing: a case study on pregnancy health and postpartum depression. *Frontiers in Psychiatry* 2023, 14, 1258887
- **Sun Z**, *et al.* A scoping review on multimodal deep learning in biomedical images and texts. *Journal of Biomedical Informatics* 2023, 104482
- Tang L, **Sun Z**, *et al.* Evaluating large language models on medical evidence summarization. *NPJ Digital Medicine* 2023, 6 (1), 158
- Wang S, Dang Y, **Sun Z** *et al.* An NLP Approach to Identify SDoH-related Circumstance and Suicide Crisis from Death Investigation Narratives. *Journal of the American Medical Informatics Association* 2023, 30 (8), 1408-1417
- **Sun Z**, *et al.* Evaluating GPT-4 on Impressions Generation in Radiology Reports. *Radiology* 2023, 307 (5), e231259
- Wang C, **Sun Z**, *et al.* Maintaining manganese in tumor to activate cGAS-STING pathway evokes a robust abscopal anti-tumor effect. *Journal of Controlled Release* 2021, 331, 480-490
- Wang C, Dong H, Liu H, **Sun Z**, *et al.* Using HPLC to analyze (S)-oxiracetam and four related substances in the bulk drug of (S)-oxiracetam. *Journal of Pharmaceutical and Biomedical Analysis* 2020, 180, 113072
- **Sun Z**, *et al.* Hydrogel-based controlled drug delivery for cancer treatment: a review. *Molecular Pharmaceutics* 2020, 17 (2), 373-391

Conference Papers

- Zhang T, **Sun Z**, *et al.* UW-BioNLP at ChemoTimelines 2025: Thinking, Fine-Tuning, and Dictionary-Enhanced LLM Systems for Chemotherapy Timeline Extraction. In *Proceedings of the 7th Clinical Natural Language Processing Workshop* 2025
- Abacha AB, Yim WW, Fu Y, **Sun Z**, *et al.* MEDEC: A Benchmark for Medical Error Detection and Correction in Clinical Notes. In *Findings of the Association for Computational Linguistics (ACL)* 2025 (pp. 22539-22550)
- Yim WW, Fu Y, **Sun Z**, *et al.* Dermavqa: A multilingual visual question answering dataset for dermatology. In *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)* 2024 (pp. 209-219)
- Yim WW, Abacha AB, Fu Y, **Sun Z**, *et al.* Overview of the MEDIQA-M3G 2024 Shared Task on Multilingual Multimodal Medical Answer Generation. In *Proceedings of the 6th Clinical Natural Language Processing Workshop* 2024 (pp. 581-589)
- Abacha AB, Yim WW, Fu Y, **Sun Z**, *et al.* Overview of the mediqa-corr 2024 shared task on medical error detection and correction. In *Proceedings of the 6th Clinical Natural Language Processing Workshop* 2024 (pp. 596-603)
- Yim WW, Abacha AB, Fu Y, **Sun Z**, *et al.* Overview of the MEDIQA-magic task at ImageCLEF 2024: multimodal and generative telemedicine in dermatology. *CLEF* 2024

Preprint

- Park N, Ahmed F, **Sun Z**, *et al.* Automated Identification of Incidentalomas Requiring Follow-Up: A Multi-Anatomy Evaluation of LLM-Based and Supervised Approaches. *arXiv preprint* 2025

Posters and Presentations

- **Sun Z**, *et al.* An NLP-Method to Identify Macro Guideline Sentences in Radiology Reports. *AMIA 2025 Annual Symposium*, Atlanta, 2025
- **Sun Z**, *et al.* An Ontology-Guided Analysis to Unveil Disparities of Social History Documentation in Discharge Notes. *AMIA 2024 Annual Symposium*, San Francisco, 2024
- **Sun Z**, *et al.* An Enhanced Multimodal Multilingual Dataset for Medical Misinformation Detection. *The 12th IEEE International Conference on Healthcare Informatics*, Orlando, 2024
- **Sun Z**, *et al.* Innovative Analysis of Social Determinants to Enhance Suicide Prevention for Black Youth. *AMIA 2023 Annual Symposium*, New Orleans, 2023

SERVICES

- **Journal Reviewers:** Journal of Biomedical Informatics (JBI), Journal of Medical Internet Research (JMIR), Biomedical Signal Processing and Control (BSPC)
- **Conference Reviewers:** Association for Computational Linguistics (ACL), American Medical Informatics Association (AMIA), IEEE International Conference on Healthcare Informatics (IEEE ICHI)
- **Membership:** Member of AMIA, Member of IEEE

SPECIALIZED SKILLS AND CERTIFICATIONS

- **Technical Skills:** Python, R, SQL, Tableau, MATLAB, GraphPad, Origin, & MS Office (Access, Excel, Outlook, PowerPoint, Word)
- **Research Skills:** Natural Language Processing, Large Language Models, Machine Learning, Data Analysis, Data Visualization, Literature Reviews, Statistical Analysis
- **Licenses & Certifications:** Google Data Analytics Specialization, IBM Data Science Specialization